An Analytical Approach Towards the Prediction of Undefined Parameters for The National Institutional Ranking Framework

Anuva Goyal
Dept. of Electrical Engineering
Faculty of Engineering
Dayalbagh Educational Institute
Agra, India
anuva.goyal111@gmail.com

Kritika
Dept. of Electrical Engineering
Faculty of Engineering
Dayalbagh Educational Institute
Agra, India
kritika4142@gmail.com

Prem Prakash Vuppuluri
Dept. of Electrical Engineering
Faculty of Engineering
Dayalbagh Educational Institute
Agra, India
vpremprakash@dei.ac.in

G.S. Sailesh Babu
Dept. of Electrical Engineering
Faculty of Engineering
Dayalbagh Educational Institute
Agra, India
gssaileshbabu@dei.ac.in

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***Abstract*—The National Institutional Ranking Framework is a methodology adopted by the Ministry of Education, Government of India for ranking Higher Education Institutions (HEIs) in India. The NIRF ranks HEIs based on five key parameters - Teaching, Learning & Resources, Research and Professional Practice, Graduation Outcome, Outreach & Inclusivity, and Peer Perception. These parameters are each composed of multiple components that are calculated based on the data provided by the respective HEIs. While the NIRF provides well-defined functions for some of these components, others are undefined. Understanding how these functions are computed would assist stakeholders in more effective planning and decision-making for further improving their NIRF scores. This work studies various regression machine learning models and identifies the best-fitting model for approximating these undefined functions. All experimental work is performed on real NIRF data publicly available on the NIRF website.**

Keywords—NIRF, MSE, Function approximation, regression

I. Introduction

The National Institutional Ranking Framework (NIRF), launched by the erstwhile Ministry of Human Resource Development (MHRD), now known as the Ministry of Education (MoE), Government of India, in September 2015, is widely used today for ranking Higher Education Institutes (HEIs) across the nation. A key motivation for the establishment of the NIRF, despite the existence of several older, well-established university ranking systems such as QS and THE World University Rankings, which have been around for some time now, is that the criteria used are, while global in scope, not necessarily suited for the Indian HEI ecosystem.

The HEIs are ranked in the NIRF using a metric based on five broad parameters as indicated in Fig.1: (a) TLR: Teaching, Learning & Resources, (b) RP: Research and Professional Practice, (c) GO: Graduation Outcome, (d) OI: Outreach & Inclusivity and (e) PR: Peer Perception. These parameters are further composed of multiple, suitably weighted components.

Fig. 1. Ranking Parameters of NIRF

The TLR parameter includes components such as the faculty-student ratio in the HEI, the number of permanent Faculty members, the number of faculty members holding PhDs, teaching, research and innovation, library, and laboratory facilities, and carries a ranking weightage of 30%. The RP parameter has a ranking weightage of 30% and represents a combined measure of the number of publications, patents, and citations. The GO parameter with a weightage of 20%, measures the outreach footprint of the institute, including components such as the percentage of students from other states, women students, as well as economically backward and socially disadvantaged students. The fourth parameter OI considers the performance of students in university examinations, their placements, the number/proportion of students going for higher studies, and the mean salaries offered to gainfully employ graduating students. Lastly, the fifth parameter, PR, is measured using

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components such as peer rating and application-to-seat ratio. The ranking weightage of the last two parameters is 10% each.

The individual HEIs are responsible for compiling and submitting the data required for evaluating these components and hence computing the weighted score for each parameter. The total score (on a maximum of 100 points), computed as the weighted sum of the individual scores for each parameter, is then used for ranking the HEIs.

For effective academic and administrative planning in an HEI, a good understanding of the relative importance of different parameters and knowledge of the precise manner in which various components are computed to obtain the parameter scores is essential. However, there are several components whose precise function is not defined in the NIRF documentation. For instance, the contribution of IPRs and patents under the RP parameter is computed as the sum of 10 x f_1(number of patents granted) + 5 x f_2(number of patents published), where the functions f_1 and f_2 are themselves not defined. This paper studies and applies different regression machine learning approaches for the purpose of approximating all such functions in the NIRF as accurately as possible.

The objective of the paper is to make a comparative study of the efficacy of different machine learning regression models for undefined parameters. To this end, a real-world dataset comprising three years' data of 186 Indian engineering institutions has been created from the public data available on the official NIRF website, and used in the experimental work presented in the paper.

The rest of the paper is organized as follows. An overview of relevant research on the subject is presented in Section II. The function approximation workflow developed in this work is described in Section III. Dataset collection, pre-processing, and organization steps are detailed in Section IV. Experimental studies on models' performance, visualization and optimization, and corresponding evaluation results are presented in Section V. Conclusions are made in Section VI.

II. Literature Review

Vikash, et al. [1] use NIRF data of IITs and NITs for the years 2016 - 2018 to correlate their rankings for various parameters defined by the NIRF. The study points out that the cumulative score of the parameters TLR, RPC, GO, OI, and PR determine the rankings. NIRF keeps altering the marks and weightage allotted to components specified under mentioned parameters, which has made it difficult to recognize the pattern of ranking of HEIs over successive years.

B. Mukherjee [2] points out that relying solely on numerical data for academic ranking can sometimes yield inaccurate results. The paper references the work of Bougnol et al. [3], which highlights the following perspective: When comparing two entities, X and Y, having nearly identical values for all attributes except one input, X is considered more efficient than Y if the value of this particular input is smaller. A potential pitfall arises when both inputs and outcomes in a ranking system are treated alike, and assigned positive weights. Consequently, a strict efficiency standpoint leads to the problem that improving rankings can be achieved by increasing the values of input attributes. To mitigate these problems stemming from input measures in a ranking scheme, one approach is to refrain from incorporating them altogether.

According to N. Doshi et al. [4], after conducting a correlation heatmap analysis on the NIRF ranking data, deduced that enhancing or altering one parameter does not necessarily lead to a similar impact on another parameter. Consequently, there appears to be a weak correlation between these parameters. The authors propose that to achieve overall rank improvement, a Higher Education Institution (HEI) should carefully examine the general ranking trends and strategize the combination of various parameters.

M.G. Ali [5] emphasizes adding more parameters such as funding and financing, including income through commercialization and licensing, Spin-off and Start-up, average tuition fees per academic year, consideration of international faculty, international collaborations, etc., along with the proper allocation of weightage to make the NIRF more robust and stable.

III. Workflow

Function approximation is a technique employed for estimating an unknown underlying function using available observations from the domain. In supervised learning, a dataset consists of inputs and outputs, and the supervised learning algorithm learns how to map examples of these inputs to respective outputs.

This mapping is ruled by a mathematical function called mapping function that a supervised learning algorithm seeks to best approximate. The optimum accuracy for this function is measured by calculating the error between the predicted values and expected values and minimizing this error by optimizing and retraining the model.

This work intends to calculate the approximate functions for NIRF parameters undefined by MHRD. For this, several function approximation methods are employed for the following components: Student Strength including Doctoral Students (SS), Financial Resources and their Utilization (FRU), Footprint of Projects, Professional Practice and Executive Development Programs (FPPP), Median Salary (GMS), Metric for Number of Ph.D. Students Graduated (GPHD) and Economically and Socially Challenged Students (ESCS).

In addition to this, the undefined functions for components Combined metric for Publications (PU), the Combined metric for Quality of Publications (QP), IPR and Patents: Patents Published and Granted (IPR), and Peer Perception: Employers & Academic Peer (PR), the data is obtained through third party sources and surveys. Hence, these components are not included in this study.

The dataset for this work has been compiled from the data provided by HEIs to the MHRD for evaluation. Further, to approximate each component's function, the extracted data has been transformed as per the component dependencies mentioned by the NIRF. For instance, the undefined function for SS under the TLR parameter requires data for the total sanctioned approved intake in the institute (NT), the total number of students enrolled in the institute

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(NE), as well as the number of students (total) who enrolled for the Ph.D. program offered by the institute up to the previous academic year (NP). The raw data for SS was transformed into a suitable dataset containing the mentioned dependencies for approximating the function.

The regression models have been trained on these derived datasets for each separately to fit the best combination of weights and bias to minimize the mean squared error. These models are then optimized to retrieve the best hyperparameters for various regression models applied. Finally, the models are compared and evaluated using MSE and ranked accordingly.

The overall workflow of the process discussed is illustrated in Fig.2.

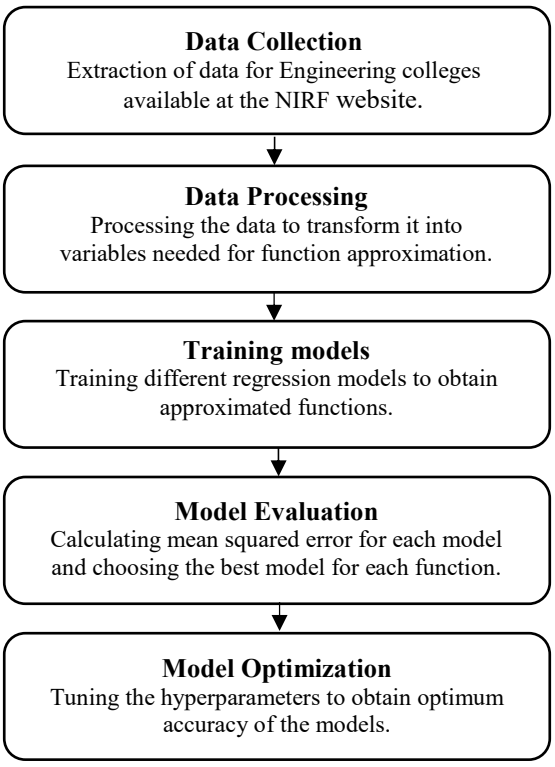

Fig 2. General workflow for function approximation

IV. About the Dataset

A. Data Collection

Data for Engineering institutions is taken from the 2022 rankings available as pdf files on the NIRF website and comprises of several tables pertaining to data of various undergraduate, postgraduate, and Ph.D. programs. This task was accomplished using the tabula.py Python library.

The pdfs were available for the following sets of academic

1. 148 files - containing 2 academic programs
   UG (4 Years), PG (2 Years)'
2. 8 files - containing 3 academic programs
   UG (2 Years), UG (5 Years), PG (2 Years)
   UG (2 Years), PG (2 Years), PG (3 Years)
   UG (2 Years), PG (2 Years), PG-Integrated
3. 6 files - containing 1 academic program
   UG (4 Years)
   PG (2 Years)
4. 4 files - containing 4 academic programs
   UG (2 Years), UG (5 Years), PG (2 Years), PG-Integrated
   UG (2 Years), PG (2 Years), PG (3 Years), PG-Integrated
   UG (2 Years), UG (5 Years), PG (2 Years), PG (3 Years)
5. 2 files - containing 5 academic programs
   UG (4 Years), UG (5 Years), PG (2 Years), PG (3 Years), PG (6 Years)
   UG (4 Years), PG (2 Years), PG (3 Years), PG-Integrated, PG (6 Years)

The first two categories have been extracted, which amounts to the data of 186 institutes. 14 files were removed as outliers. In addition, scores of each component have been extracted for the corresponding institutes. The complete dataset is made available [1] for access to interested researchers.

CollegeID	UG4_MaleStudent	UG4_FemaleStudent	UG4_TotalStudents
IR-E-U-0255.pc	454	33	487
IR-E-U-0184.pc	706	171	877
IR-E-U-0355.pc	973	185	1158
IR-E-U-0205.pc	2889	492	3381
IR-E-U-0701.pc	3157	548	3705

CollegeID	Total_Phd_FT	Total_Phd_PT	Graduate_FT_20_21	Graduate_FT_19_20
IR-E-U-025	176	54	12	13
IR-E-U-018	286	19	52	48
IR-E-U-035	288	0	35	35
IR-E-U-020	502	391	76	68
IR-E-U-070	1152	6	112	115
IR-E-U-001	38	255	8	5

Fig. 3. Snippets of raw data

B. Data Preprocessing

The collected raw data has been processed to convert it into an efficient format. Some calculations have been performed on the raw data to obtain the datasets required as per each component to train the model.

CollegeID	NESC	ESCS	BC	BO	FRU
IR-E-C-1412	0.842001	4.67	14782.32	133113.6	7.82
IR-E-U-0064	34.50586	5.22	165558	390246.1	18.73
IR-E-C-1660	15.92602	3.96	11471.4	227330.8	11.13
IR-E-U-0149	11.10422	3.28	23119.46	255449.7	11.74
IR-E-U-0197	66.97971	11.2	92002.51	262275.7	14.59
IR-E-I-1356.	2.842377	2.95	7342.041	65136.52	4.28
IR-E-I-1480.	4.910486	0.79	39204.79	251505.2	12.73
IR-E-U-0613	44.39252	8.94	62250.31	205711.4	11.03
IR-E-U-0055	44.63109	7.31	50341.2	221595.5	12.58
			7320.043	74465.7	4.87

Fig. 4. Snippets of pre-processed data

[1] https://github.com/Kritika-4142/NIRF_Dataset

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College ID has been used as a primary key to map the fields from various raw data files, as shown in Fig. 3 to form the processed dataset for each component which is shown in Fig. 4.

V. Training, Optimizing, and Evaluating Models

A. Training Regression Models

Regression analysis is a statistical approach utilized to establish a connection between dependent (target) and independent (predictor) variables, involving one or more independent variables. The primary objective of regression is to minimize the difference between the predicted values and the actual values of the target variable. This is typically achieved by fitting the regression model to a training dataset and optimizing the model's parameters to reduce the prediction error. Once the model is trained, it can be used to make predictions on new, unseen data.

The machine learning models employed for function approximation are Linear Regression, Polynomial Regression, KNN Regression, Support Vector Regression, Decision Tree Regression, Random Forest Regression, Convolutional Neural Networks, and XG Boost.

The models are trained on a Kaggle environment with disk space of up to 73.1 GB and RAM of up to 30 GB.

B. Model Optimization

Model optimization is the process of modifying hyperparameters to minimize the cost function by using an optimization technique. It is essential to reduce the cost function because it describes the discrepancy between the actual value and the predicted value of the estimated parameter. To achieve this objective, the scikit-optimize library's BayesSearchCV is employed. BayesSearchCV comes equipped with essential methods such as "fit" and "score." Additionally, if the estimator used for optimization supports them, BayesSearchCV also provides implementations for "predict," "predict_proba," "decision_function," "transform," and "inverse_transform." Furthermore, the parameter settings of the estimator are optimized through cross-validated search to enhance the performance.

C. Visualizing model performances

The actual data points and the predicted data points are plotted to visualize and analyze the models' performances.

For the succeeding graphs, the scatter plot depicts the actual data points while the line plot depicts the predicted or obtained data points.

1. **For TLR_SS (20 Marks)**

- $SS = f(NT, NE) \times 15 + f(NP) \times 5$
- The functions f (NT, NE) and f(Np) are determined by NIRF.
- NT represents the total sanctioned approved intake in the institution, including all undergraduate (UG) and postgraduate (PG) programs.
- NE represents the total number of students enrolled in the institution, encompassing all UG and PG programs.
- Np represents the total number of students enrolled in the doctoral program up until the previous academic year.

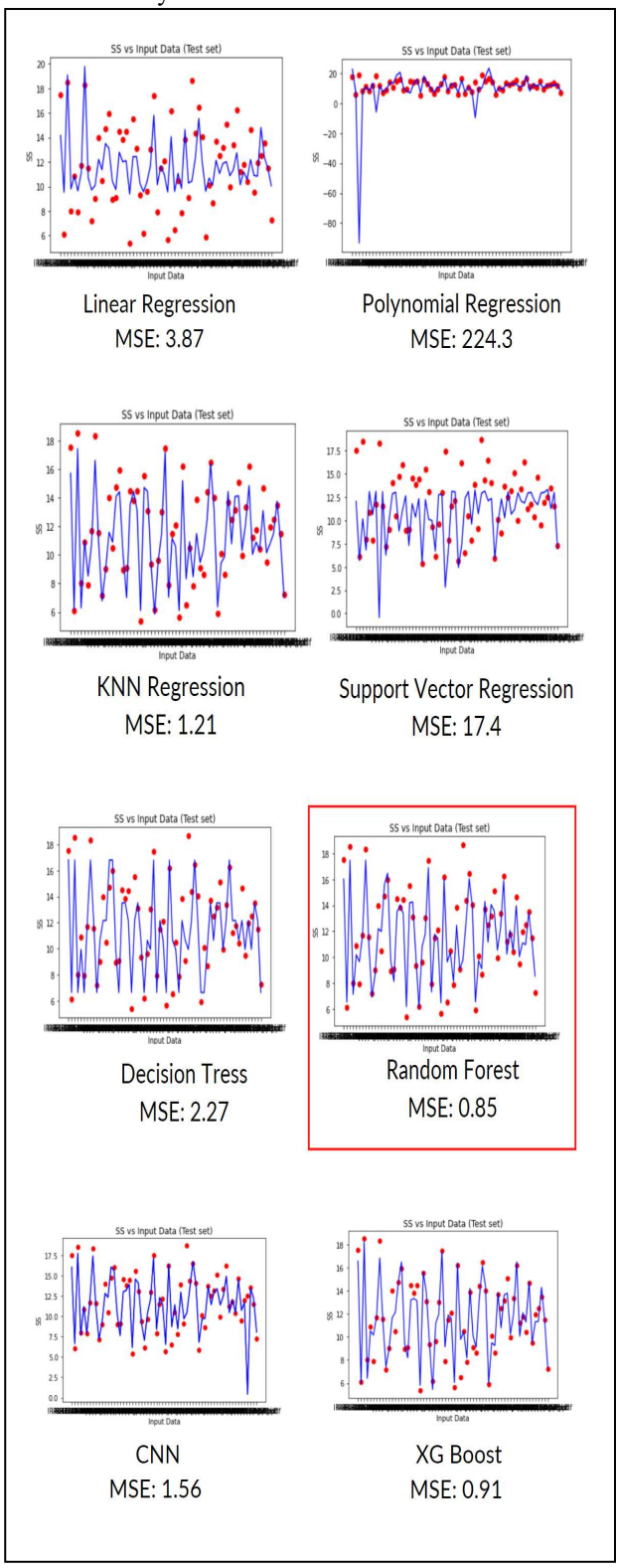

Fig. 5. Visualizing Regression model predictions for parameter SS

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2. **For GO_GPHD (20 Marks)**

- $GPHD = 20 \times f(Nphd)$
- Nphd represents the average count of Ph.D. students who successfully completed their doctoral degrees over the preceding three years.

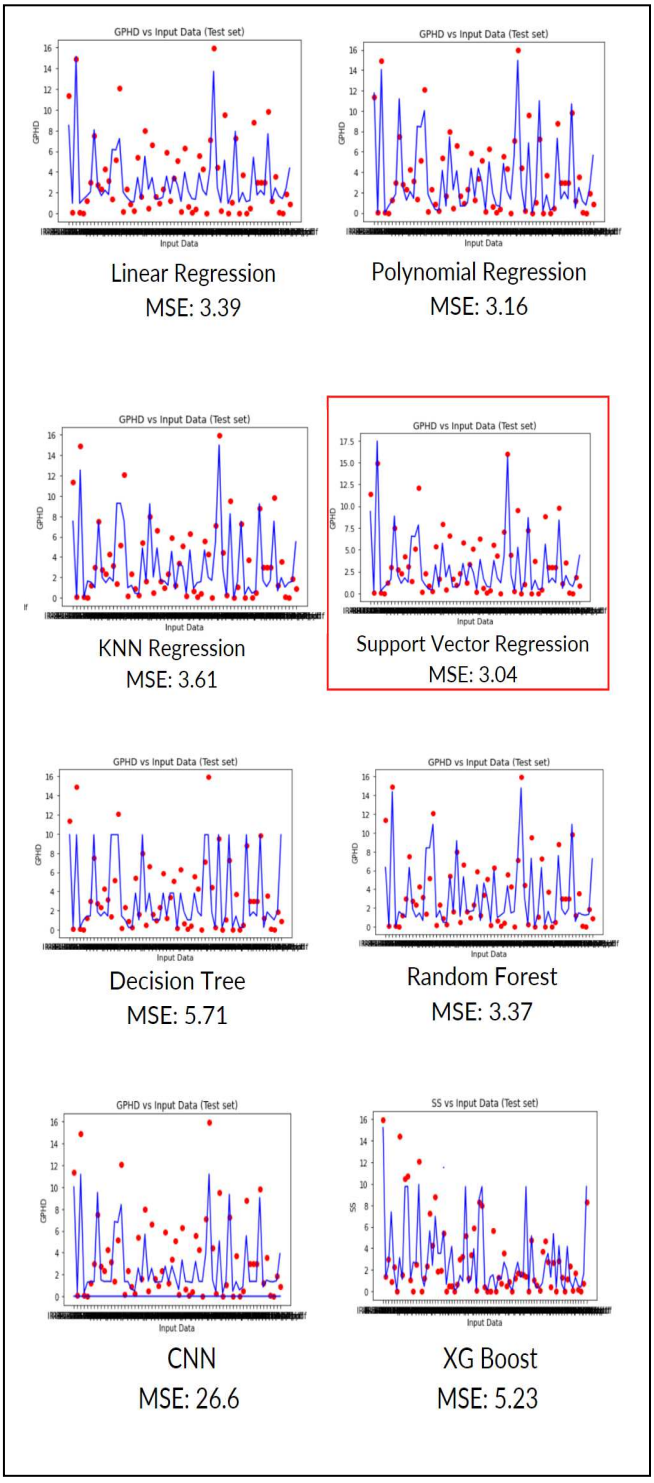

Fig. 6. Visualizing Regression model predictions for parameter GPHD

3. **For GO_GMS (25 Marks)**

- $GMS = 25 \times f(MS)$
- MS represents the median salary earned by graduates (both from UG and PG programs) in the institution during the preceding three years.

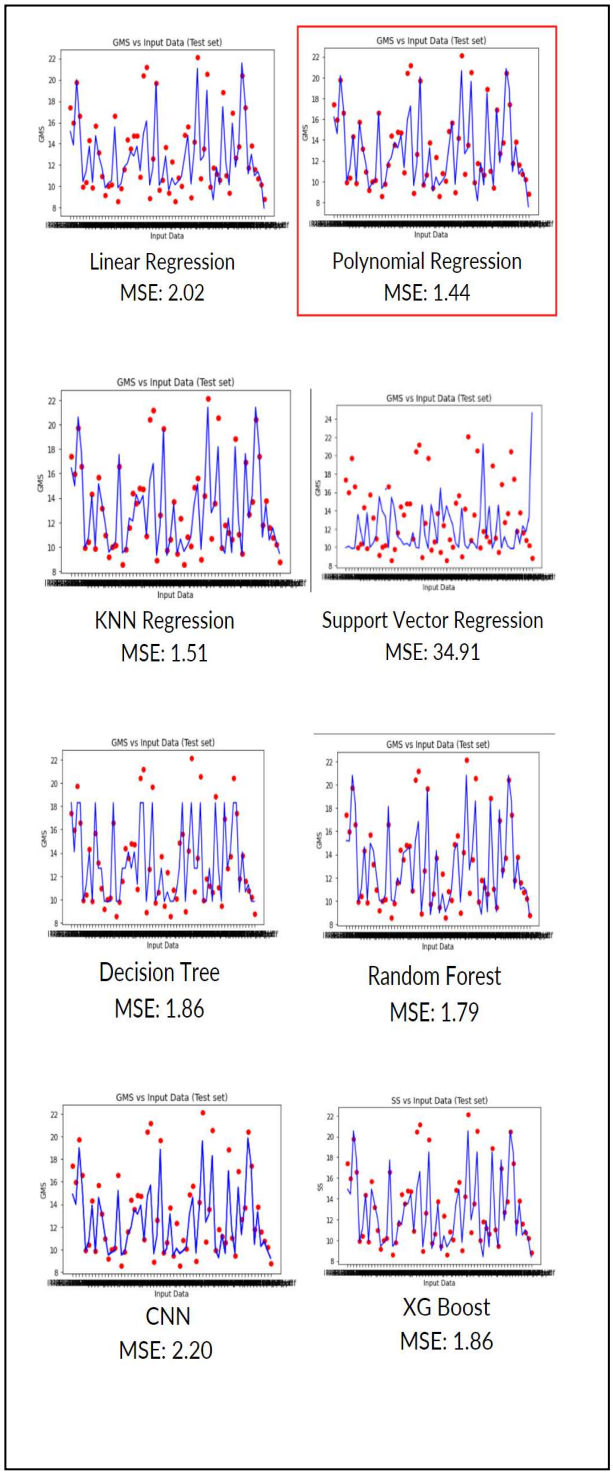

Fig. 7. Visualizing Regression model predictions for parameter GMS

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4. **For OI_ESCS (20 marks)**
- ESCS = 20 × f(Nesc)
- Nesc represents the percentage of UG students who receive full tuition fee reimbursement from the institution to support their degree programs.

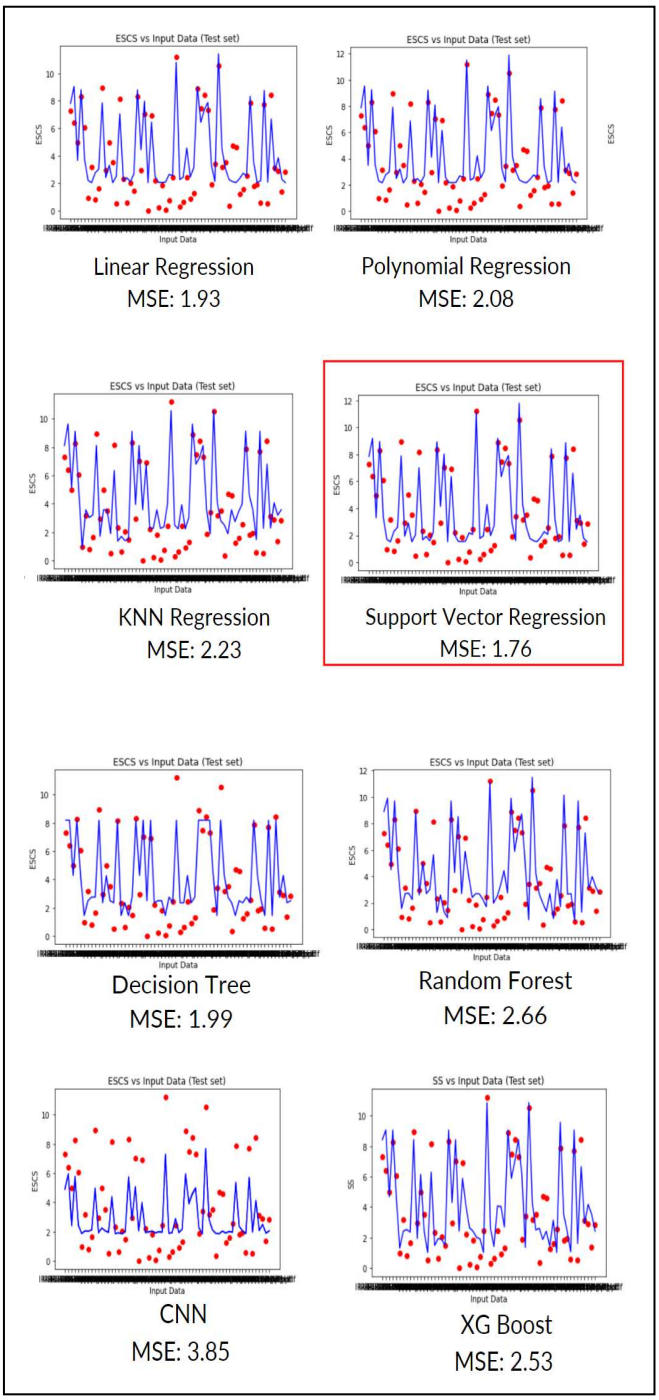

Fig. 8. Visualizing Regression model predictions for parameter ESCS

5. **For TLR_FRU (30 Marks)**
- FRU = 7.5 × f(BC) + 22.5 × f(BO)
- BC represents the average annual capital expenditure per student for the past three years, specifically related to the engineering discipline. This excludes any expenditure on the construction of new buildings.
- BO represents the average annual operational (or recurring) expenditure per student for the previous three years, pertaining to the engineering discipline only. This excludes expenses related to the maintenance of hostels and allied services.

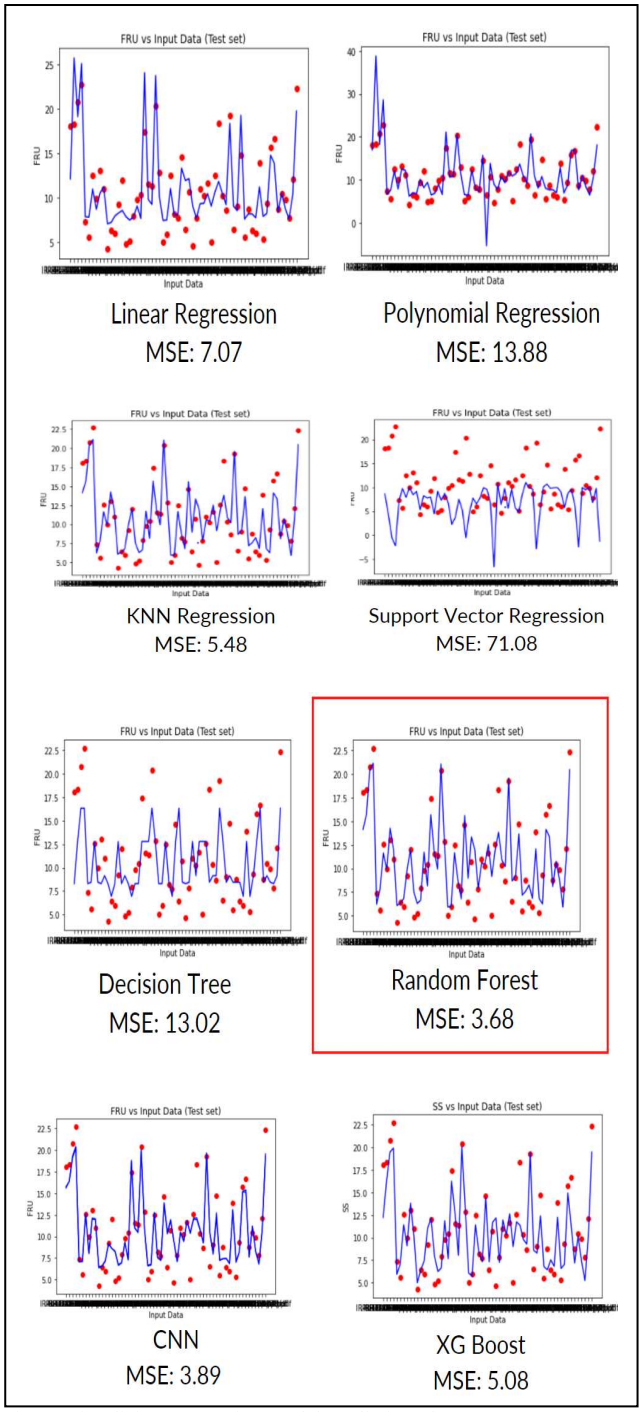

Fig. 9. Visualizing Regression model predictions for parameter FRU

6. **For RP_FPPP (10 marks)**
- FPPP = FPR + FPC
- FPR = 7.5 × f (RF)
- RF represents the average annual research funding earnings (in rupees) per faculty at the institute level over the past three years.
- FPC = 2.5 × f (CF)

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- CF represents the average annual consultancy amount (in rupees) per faculty at the institute level during the previous three years.

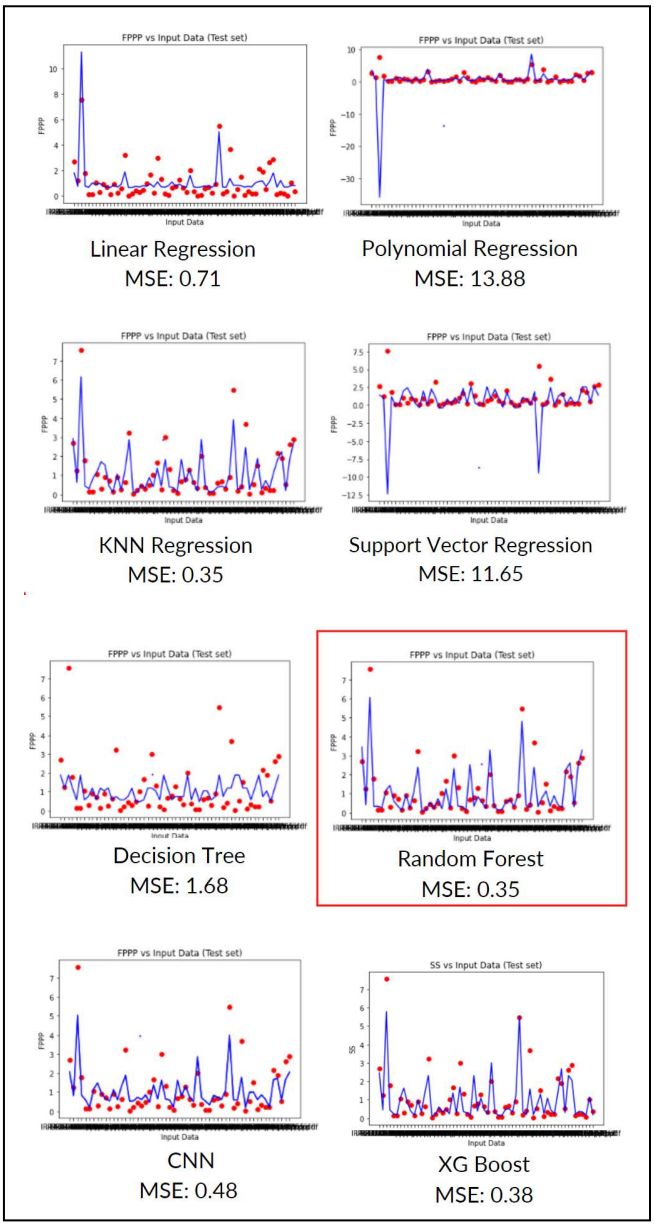

Fig. 10. Visualizing Regression model predictions for parameter FPPP

D. Model Evaluation

To evaluate the models, the Mean Squared Error (MSE), a regression metric, is employed to quantify the error. It measures the average squared discrepancy between the observed and predicted values. When a model has no error, the MSE equals zero. As model error increases, its value increases. The MSE is also known as the mean squared deviation (MSD).

The MSE in regression stands for the average squared residual. The MSE decreases as the data points get closer to the regression line because the model has less error. Predictions made by a model with lower error are more accurate.

The formula for MSE is represented by Equation 1.

$$MSE = \frac{\sum(y_n - \hat{y}_n)^2}{N} \quad (1)$$

where:
y_n = n^{th} observed value.
$\hat{y}_n$ = n^{th} predicted value.
N = total number of observations.

The MSE for the models after hyperparameter tuning is presented in Table 1.

TABLE I. MSE FOR REGRESSION MODELS FOR EACH COMPONENT

Models	SS	GPHD	GMS	ESCS	FRU	FPPP
Linear Regression	3.87	3.39	2.02	1.93	7.07	0.71
Polynomial Regression	24.3	3.16	1.44	2.08	13.88	13.88
KNN Regression	1.18	3.61	**1.37**	2.23	5.48	**0.35**
SVR	17.4	**3.04**	3.91	**1.76**	7.08	11.65
Decision Trees	1.92	5.2	1.6	1.99	5.035	0.49
Random Forest	0.85	3.37	1.4	2.25	**3.68**	**0.35**
CNN	1.56	26.6	2.2	3.85	3.89	0.48
XG Boost	**0.76**	5.23	1.86	2.29	5.08	0.38

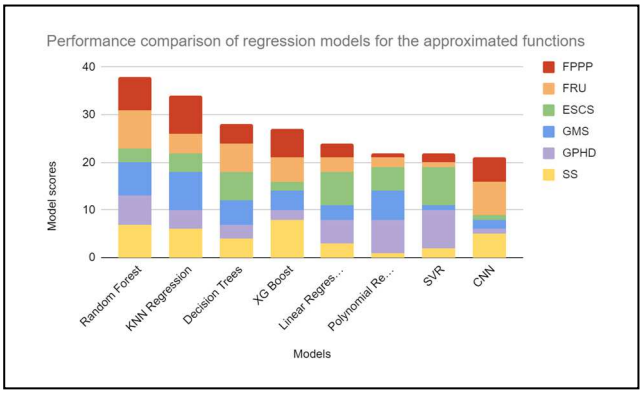

Fig. 11. Rankings of regression models as per their performance

The models are ranked from 1 to 8 as per their mean squared error,1 for the highest MSE and 8 for the least MSE. The cumulative rank was calculated, and it can be observed from Fig. 11 that Random Forest Regression is giving the optimum results for all the six approximated functions.

VI. CONCLUSIONS

This paper employs regression models to estimate the values of undefined parameters in NIRF data and presents a comparative study of the performance of different models to identify the one that yields the best possible outcomes. To the best of the authors' knowledge, this is the first attempt to utilize machine learning techniques for enhanced prediction of NIRF ranking. Another important contribution of this work is the creation of an authentic and substantive dataset sourced directly from the NIRF website, which can be used for benchmarking further research endeavors in this field.

From the experimental work described, it is observed that the Random Forest regression model exhibits the best

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performance out of the various machine learning models considered in the study. Since the Random Forest technique is based on trees, the scaling of the variables does not matter in this approach. Moreover, Random Forest is an ensemble learning algorithm that combines multiple decision trees to make a final prediction. Each decision tree is trained on a random subset of the data, and the final prediction is made by taking the majority vote of all the trees. This ensemble approach helps to reduce the variance and overfitting that can occur when using a single decision tree. Random Forest has performed well because of the following reasons:

1. Feature selection: Random Forest selects a random subset of features for each decision tree. This helps to reduce the impact of irrelevant or noisy features and can improve the accuracy of the model.
2. Non-linear relationships: Random Forest can capture non-linear relationships between the features and the target variable. This is because each decision tree can model different interactions between the features, and the ensemble approach can combine these interactions to make a final prediction.
3. Robustness: Random Forest is a robust algorithm that can handle missing values and noisy data. It also performs well with imbalanced datasets, where the number of samples in each class is not equal.

In general, the Random Forest method is a versatile and powerful algorithm that can be used for a wide range of classification and regression tasks. Its high accuracy, robustness, and scalability make it a popular choice in machine-learning applications, as is borne out in this work.

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